

# Java 8 Features

**Purpose:** A thorough, academic-oriented guide for interns to learn and implement key Java 8 features: Lambda expressions, Method references, Functional interfaces, Default & Static methods in interfaces, Marker interfaces, Stream API, forEach, Collectors, and practical guidance for defining and implementing interfaces.

**Prerequisites:** JDK 8 or later, basic Java (classes, objects, collections), an IDE (IntelliJ/Eclipse/VS Code) and familiarity with generics and JVM basics.

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## Learning objectives

- Java 8 functional programming concepts and why they were added to the language.
- Write and read lambda expressions and method references responsively.
- Create and consume Streams and use collectors for aggregation and grouping.
- Define functional interfaces, add default and static methods to interfaces, and understand method resolution rules.
- Build clear, side-effect-free stream pipelines and reason about parallelism.
- Implement small projects using these features in idiomatic Java 8 style.